# Unit 1: Lesson 2 **Grade Level:** Middle School

# Input & Output

## Description

Students will understand how computers rely on input and output to function by exploring real-world examples and designing their own I/O system using the Input → Process → Output model.

## Objectives

1. Identify common input and output devices and describe their purpose.
2. Explain the Input → Process → Output (IPO) model and why it's essential for computers to function.
3. Analyze real-world systems that rely on input and output to solve everyday problems.
4. Design a creative I/O system by identifying appropriate input and output devices that address a real-world challenge.
5. Describe their designed system clearly, explaining how the input, process, and output interact.
6. Reflect on the importance of input/output in modern technology and identify I/O systems they use in daily life.

## Time Frame: 1, 50 min class

## Materials

- A computer with access to the internet
- Printable or digital Input/ Output Cards
- Printable or digital Brainstorm Sheet
- Visual aid for the IPO Model

## Lesson Procedures

## Anticipatory Set (5-10 mins)

**Activity: "The Silent Computer" Demo**
Show a brief demonstration that emphasizes the importance of Input/Output:

- Input Failure: Type on a keyboard with the monitor turned off → No result.
- Output Failure: Speak into a muted microphone → No sound.
- Both Missing: Power on a computer with no devices connected → Nothing happens.

**Guiding Question:** *"If computers can't send or receive information, what's their purpose?"*
**Conclusion:** Emphasize that I/O is what gives computers purpose — without it, they can't respond or interact with the world.

## Guided Practice: 10 mins

**Activity: "Input/Output Match-Up"**
Provided in the resources document of this lesson are pdfs that have an image on them. Each represents an input and an output. The goal is to have the students match the input and output correctly to understand the relationships between devices. You may also make your own, using the ones available as a guide. Try to incorporate objects that are not computer devices that also have an input/output.

Complete this lesson by having the students:

- Using a projector, smart screen, or sheets of paper; have the students come up 1 by 1 and match the input to the output on the screen.
- Print out enough for small groups and have the students collaborate with each other to come up with the answers.
- Create a Google form and have them match on their own.

### Class Discussion

- *"Which pairs were surprising?"*
- *"What would happen if the input or output in your system stopped working?"*

## Project-Based Learning (25 mins)

**Activity: "Design Your Own I/O System"**

Students will be brainstorming their own I/O systems that can solve problems that exist in the classroom or in the real world. The idea for this assignment is for the students to examine what input and output mean on a small or large scale.

Students will be given a brainstorming sheet that asks the following questions:

- What's the problem you're solving?
- What's the input device? (e.g., Button, sensor, etc.)
- What's the output device? (e.g., Light, sound, etc.)
- How does your system help people?
- Illustrate your system with a labeled diagram or write a short explanation describing how their system works.

### Sample Solutions

Provide the students with examples of what you're looking for. Here are a few to help you get started. I encourage you to come up with some on your own that make sense for your class.

- **Automatic Desk Light** that detects lowlight. It turns on automatically when the light is too low.
- **Backpack Reminder System**. Its a sensor that activates when you leave the class to remind you to grab your backpack. The speaker says, "Did you grab your backpack?"
- **Classroom Timer**. Used for any task that needs a timer. The input (button) starts a countdown and flashes when the timer is up.

## Closure/Reflection: 5 mins

Remember to reiterate what the students have learned. Have students share their systems. Here are some sample reflection questions to ask:

- *"What did you learn about how computers rely on input/output?"*
- *"What would happen if your system's input or output failed?"*

### Key Message for Students

*"Computers are powerful problem solvers because they can take in information (input) and respond with useful results (output). Without I/O, computers wouldn't be able to help us in everyday life."*

### California/National Computer Science Standards

This lesson meets the following standards set by the state and board of education. 6-8.CS.1, 6-8.CS.2, 6-8.AP.10, 6-8.AP.11, 6-8.IC.20

### Adaptations for Diverse Learners

**Visual Support:** Provide a visual IPO Model diagram and include picture-based matching cards during the Input/Output Matching Activity to support visual learners and ELL students.

**Collaborative Learning:** Pair students for both the Matching Activity and Design Challenge, allowing students to discuss ideas aloud and support each other's understanding.

### Criteria for Assessment

*Students will be assessed on their ability to:*

- Accurately identify input and output devices in the Matching Activity.
- Clearly describe the input, process, and output in their designed I/O system.
- Present a creative or practical solution that effectively addresses a real-world problem.
- Communicate their ideas clearly through writing, diagrams, or verbal explanation.

### Methods for Assessment

- **Observation:** Monitor student participation during the Matching Activity and Design Challenge for engagement and understanding.
- **Checklist/Rubric:** Use a simple checklist to evaluate the accuracy and completeness of each student's I/O system design.
- **Student Reflection:** Collect student responses to the exit prompt: "Describe an input/output system you use every day and explain why it's important."

 